Int J Syst Assur Eng Manag
https://doi.org/10.1007/s13198-024-02440-6

ORIGINAL ARTICLE

Addressing data imbalance challenges in oral cavity histopathological whole slide images with advanced deep learning techniques

Tabasum Majeed[1] · Tariq Ahmad Masoodi[2] · Muzafar Ahmad Macha[3] · Muzafar Rasool Bhat[4] · Khalid Muzaffar[5] · Assif Assad[1]

Received: 18 November 2023 / Revised: 9 May 2024 / Accepted: 7 July 2024
© The Author(s) under exclusive licence to The Society for Reliability Engineering, Quality and Operations Management (SREQOM), India and The Division of Operation and Maintenance, Lulea University of Technology, Sweden 2024

Abstract Oral Cavity Squamous Cell Carcinoma (OCSCC) represents a common form of head and neck cancer originating from the mucosal lining of the oral cavity, often detected in advanced stages. Traditional detection methods rely on analyzing hematoxylin and eosin (H&E)-stained histopathological whole-slide images, which are time-consuming and require expert pathology skills. Hence, automated analysis is urgently needed to expedite diagnosis and improve patient outcomes. Deep learning, through automated feature extraction, offers a promising avenue for capturing high-level abstract features with greater accuracy than traditional methods. However, the imbalance in class distribution within datasets significantly affects the performance of deep learning models during training, necessitating specialized approaches. To address the issue, various methods have been proposed at both data and algorithmic levels. This study investigates strategies to mitigate class imbalance by employing a publicly available OCSCC imbalance dataset. We evaluated undersampling methods (Near Miss, Edited Nearest Neighbors) and oversampling techniques (SMOTE, Deep SMOTE, ADASYN) integrated with transfer learning across different imbalance ratios (0.1, 0.15, 0.20, 0.30). Our findings demonstrate the effectiveness of SMOTE in improving test performance, highlighting the efficacy of strategic oversampling combined with transfer learning in classifying imbalanced medical datasets. This enhances OCSCC diagnostic accuracy, streamlines clinical decisions, and reduces reliance on costly histopathological tests.

Keywords Imbalanced data · Deep learning · Histopathology · Whole slide Images · Oral cavity squamous cell carcinoma · Oversampling · Undersampling

1 Introduction

Oral squamous cell carcinoma (OSCC), which is the predominant form of cancer in the head and neck region originates

✉ Assif Assad
assif.assad@islamicuniversity.edu.in

Tabasum Majeed
tabasum.majeed@iust.ac.in

Tariq Ahmad Masoodi
tahamasoodi@gmail.com

Muzafar Ahmad Macha
muzafar.macha@iust.ac.in

Muzafar Rasool Bhat
muzafar.rasool@islamicuniversity.edu.in

Khalid Muzaffar
khalid.muzaffar@islamicuniversity.edu.in

[1] Department of Computer Science and Engineering, Islamic University of Science and Technology, Awantipora, Kashmir, India

[2] Department of Cancer Research, Sidra Medicine, Doha, Qatar

[3] Watson-Crick Centre for Molecular Medicine, Islamic University of Science and Technology, Awantipora, Kashmir, India

[4] Department of computer science, Islamic University of Science and Technology, Awantipora, Kashmir, India

[5] Department of Electronics and Communication Engineering, Islamic University of Science and Technology, Awantipora, Kashmir, India

Published online: 26 July 2024

specifically from the mucosal lining of the oral cavity (Du et al. 2020; Li et al. 2022). It accounts for over 90% of all oral cancer cases (Warnakulasuriya and Greenspan 2020). The World Health Organization's data indicates that approximately 657,000 new oral cavity squamous cell carcinoma (OCSCC) cases are diagnosed yearly, resulting in more than 330,000 deaths. Significantly higher incidence rates of OCSCC have been observed in South Asian nations. India accounts for the most significant proportion of these cases, comprising one-third of the total, while Pakistan has the highest incidence of this cancer in males and females, ranking first and second, respectively (Anwar et al. 2020). Detecting OCSCC in its early stages is crucial for achieving successful treatment, improving survival rates, and reducing mortality and morbidity (Chakraborty et al. 2019).

The diagnosis of oral cavity squamous cell carcinoma generally requires a range of procedures, encompassing physical examination, various imaging tests like digital mammograms (X-rays), MRI (Magnetic Resonance Imaging), ultrasound, and biopsies, along with occasional testing for Human papillomavirus (HPV), to determine and verify the presence of cancer accurately. However, a biopsy is the most reliable way to diagnose oral cavity squamous cell carcinoma. This procedure involves collecting and extracting cell or tissue samples for microscopic analysis. Pathologists then thoroughly examine these histopathology samples under a microscope, typically using Hematoxylin-Eosin (H&E) staining. This examination focuses on specific areas of the whole-slide tissue scans, evaluating the overall tissue structure and variations. This thorough analysis is crucial for determining the grade and stage of the disease, as well as assessing its aggressiveness.

However, this diagnostic pathology method depends on histopathologist's analysis, a process that can be time-consuming and error-prone, thereby constraining its effectiveness in clinical applications (Santana and Ferreira 2017). Consequently, there is an increasing demand for computerized automation to enhance diagnostic reproducibility across various scales (McCann et al. 2014).

Conventional cancer diagnostic methods have focused on integrating pattern recognition systems to facilitate rapid and automated detection. This process typically involves using specially selected features like texture and morphology extracted from histological images. These features are subsequently utilized in training a classifier to identify cancerous cells. The paradigm recently shifted towards deep learning techniques, wherein feature extraction and classifications are amalgamated into a unified framework. Deep Convolutional Neural Networks (Deep CNNs), which are feed-forward artificial neural networks featuring multiple hidden layers, have become ubiquitously utilized across various applications such as object tracking, detection, image classification, and computer vision, owing to their demonstrable efficacy (Liu and An 2020; Mittal et al. 2019). Deep learning (DL), especially, has demonstrated remarkable achievements in addressing various tasks related to medical image processing (Altaf et al. 2019), particularly in the field of diagnosing histopathological images (Deif et al. 2021; Echle et al. 2021). Computer-aided diagnosis (CAD) systems utilizing deep learning have been proposed and widely implemented for various types of cancer, including breast cancer (Duggento et al. 2021), prostate cancer (Goldenberg et al. 2019), and lung cancer (Wang et al. 2019). However, recent research indicates that deep learning (DL) has rarely been used to diagnose OCSCC based on histopathology images. In a research study (Das et al. 2018), the authors employed CNN and Random Forest to identify keratin pearls in oral histopathology images. In their research study, CNN achieved an accuracy of 98.05% for keratin area segmentation, while the Random Forest model detected keratin pearls with an accuracy of 96.88%. In another research study (Folmsbee et al. 2018), the authors employed a CNN to classify oral cancer tissue types into seven distinct categories. This classification included stroma, tumor, lymphocytes, mucosa, keratin pearls, adipose, and blood and integrated Active Learning (AL) and Random Learning (RL) techniques. It was observed that the AL method outperformed the RL technique, achieving an accuracy that was 3.26% higher.

Despite the numerous advantages of deep learning, it faces significant challenges, particularly concerning datasets. The effectiveness of a deep learning model is highly contingent on the quality, size, and diversity of the dataset it is trained on. Data-related issues can manifest in various forms, such as insufficient data volume, lack of variability within the data, imbalances in class representation, and issues with data quality and labeling accuracy. However, class imbalance, a widespread and crucial issue where the number of instances (data points or samples) in one class is significantly greater than in another, is frequently observed in numerous real-world applications (Choi et al. 2016; Wei et al. 2013). To address the issue of class imbalance, sophisticated techniques for oversampling and undersampling, such as the Synthetic Minority Over-sampling Technique (Chawla et al. 2002), Deep SMOTE (Dablain et al. 2022), ADASYN (He et al. 2008) (SMOTE), Edited Nearest Neighbour (ENN) (Wilson 1972), and others, are utilized. These methods operate by algorithmically enhancing the representation of minority and majority classes in the dataset. This is achieved through the generation of synthetic samples or by editing existing samples, ultimately leading to a more balanced class distribution.

However, data imbalance issues are particularly prevalent in the medical domain, where the number of normal patients often significantly exceeds those of cancerous patients. This study explicitly addresses this imbalance in a publicly available OSCC dataset. To realistically replicate

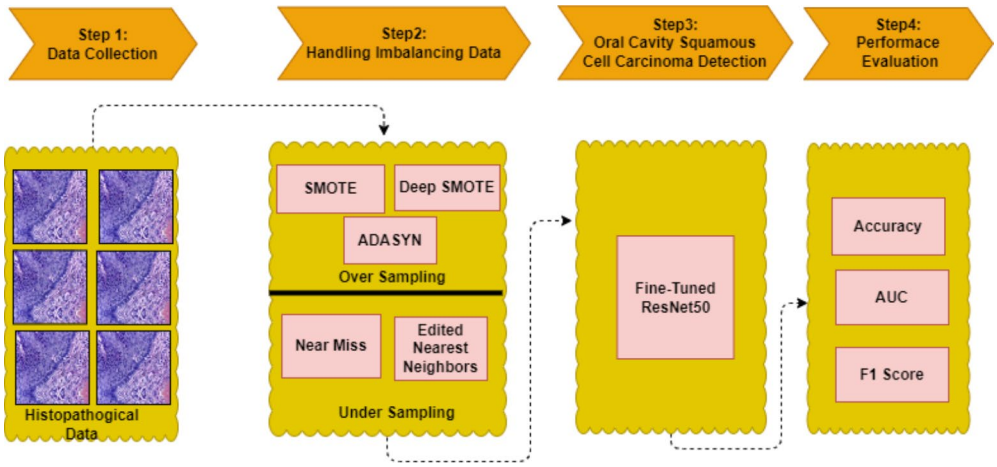

Fig. 1 Methodology adopted

these imbalanced scenarios, we incorporated different imbalance ratios, namely 0.1, 0.15, 0.2, and 0.3, within our training dataset. We then employed various oversampling and undersampling techniques to mitigate this disparity. These techniques encompassed algorithms such as the SMOTE, Deep SMOTE, Adaptive Synthetic Sampling (ADASYN), Near Miss, and ENN. The application of these techniques aimed at recalibrating the dataset to achieve a more balanced representation. This was done by either augmenting data in the minority class (in cases of oversampling) or reducing instances in the majority class (in cases of undersampling). Such adjustments are crucial to creating a training environment more reflective of diverse scenarios, enhancing the model's training and validation processes. Notably, our exploration indicated that using SMOTE with transfer learning was particularly effective in classifying imbalanced histopathology images, which holds significant promise in improving diagnostic accuracy and, consequently, patient outcomes in the medical field.

The main contributions of this paper are:

- This study presents the utilization of varying imbalance ratios in the OCSCC dataset by developing a deep learning-based pipeline to accurately simulate real-world diagnostic scenarios in medical imaging.
- We successfully demonstrate the effectiveness of advanced oversampling and undersampling techniques in addressing the class imbalance, thereby enhancing the accuracy of OCSCC detection.
- Our research contributes to the field by integrating transfer learning with SMOTE, a combination that shows significant promise in improving the classification accuracy of histopathology images in medical diagnostics.

The following sections are structured: Section 1 presents our research's background, motivation, and goals. Section 2 provides an overview of the materials and methods used in this study. Section 3 discusses the results, summarizing the key points and implications of the research.

2 Materials and methods

This research study mainly uses sampling techniques and transfer learning to detect oral cavity squamous cell carcinoma in histopathology images. To employ this distinct framework effectively, we followed a systematic approach encompassing four steps: dataset collection, dealing with Imbalanced Class, model development, and model evaluation. Figure 1 depicts the methodology adopted for detecting oral cavity squamous cell carcinoma.

2.1 Dataset collection

In our study, we use a dataset from a GitHub repository[1], which consists of H&E whole slide images of both normal oral cavity epithelium and Oral cavity Squamous Cell Carcinoma (OCSCC) cases. This dataset includes 1224 histopathological images, divided into two sets with differing resolutions. The first set has 89 images of normal oral cavity epithelium and 439 images of OCSCC, all captured at 100× magnification. The second set contains 201 images of normal oral cavity epithelium and 495 images of OCSCC, each taken at 400× magnification.

[1] https://github.com/Tabassum2019/A-histopathological-image-repository-of-normal-epithelium-of-Oral-Cavity-and-OSCC/blob/master/README.md

We selected 201 images from each class in the second set for our experimentation setup. This dataset was then divided, with 70% allocated for training and the remaining 30% for testing. Subsequently, intentional imbalance was introduced in the training dataset to investigate its impact on the accuracy of medical classification. Varying proportions of OSCC samples ranging from 10 to 30% were employed in the training dataset, resulting in the creation of four distinct imbalanced training datasets denoted as OCSCC subsets A, B, C, and D. Each subset represents a different level of class imbalance, facilitating comprehensive analysis of its effect on classification performance in a medical context. The testing set comprised fifty-nine histopathology whole slide images, consistent across all training datasets.

2.2 Dealing with imbalanced class

Addressing imbalanced data is a prevalent challenge in supervised learning, posing difficulties in training deep learning models. Usually, classification algorithms work best with data where all classes are equally represented. However, in medical settings, identifying rare cases (the minority class) is often more important than identifying common ones (the majority class). For example, in medical classification, correctly diagnosing a patient with cancer (a less common case) is usually more crucial than identifying patients without cancer. To tackle this, several methods have been developed to reduce the bias towards more common classes in imbalanced data situations. In this research study, we employed different oversampling and undersampling techniques to address the issue of imbalanced classes.

2.2.1 Undersampling

This approach involves adjusting the dataset by eliminating instances from the majority classes until the minority class reaches a certain proportion of the majority class. Additionally, the greater the imbalance in the dataset, the more samples from the majority class need to be removed during the undersampling process. In this study, we have implemented Near Miss Undersampling (Wilson 1972) and Edited Nearest Neighbors Rule (Wilson 1972) as our primary undersampling strategies for addressing the imbalance in oral cavity squamous cell carcinoma data.

The Edited Nearest Neighbors (ENN) method is employed to refine the dataset by selectively removing those data points whose class labels do not match the predominant class among their k nearest neighbors. This technique enhances the homogeneity of the local neighborhood within the dataset. Concurrently, the NearMiss (NM) methods primarily focus on the spatial distances between data points within the majority class. The fundamental idea is to select samples from the majority class that are closest in space to the minority class samples. This approach helps retain more informative or representative samples from the majority class while removing those that are less informative or redundant.

2.2.2 Oversampling

This approach addresses class imbalance in a dataset, especially when one class (the minority class) is significantly less prevalent than another (the majority class). The objective of oversampling is to augment the presence of the minority class to establish a more balanced dataset. In this study, we have implemented SMOTE (Chawla et al. 2002), Deep SMOTE (Dablain et al. 2022), and ADASYN (He et al. 2008) as our Oversampling strategies for addressing the imbalance in oral cavity squamous cell carcinoma data.

SMOTE creates synthetic samples for the minority class in a dataset by interpolating between existing minority instances and their nearest neighbors. This approach enhances class balance, improving model training and generalization without duplicating minority data. Deep SMOTE aims to address class imbalance in datasets by generating more complex and representative synthetic samples for the minority class. ADASYN (Adaptive Synthetic Sampling) addresses class imbalance by developing synthetic data, focusing more on minority class instances that are harder to classify. It adapts the number of synthetic samples based on the local density distribution of each minority instance, reducing bias and improving overall classification accuracy.

2.3 Model development

This section outlines the architecture and hyperparameter settings of the Fine-Tuned ResNet 50 model, which have been trained through transfer learning. Transfer learning is a powerful deep learning technique involving using a pre-trained model and adapting it to a new but related task.

The ResNet50 model, a convolutional neural network, is known for its deep network of layers arranged sequentially. These layers include convolutional layers for pattern recognition, pooling layers for compressing the data, and fully connected layers for making predictions. Unique to ResNet50 are residual or skip connections, which help combat the vanishing gradient problem during backpropagation. This is achieved by allowing the gradient to flow through these shortcut connections. ResNet50 is particularly noted for its performance in complex tasks, such as the ImageNet challenges (He et al. 2016). It employs 3×3 filters for efficient feature extraction and maintains a consistent computational complexity by adjusting the number of filters based on the size of the feature map. Additionally, ResNet50 effectively reduces spatial dimensions using a stride of two in its convolutional layers and incorporates a global average pooling layer for a more refined and comprehensive global representation of features.

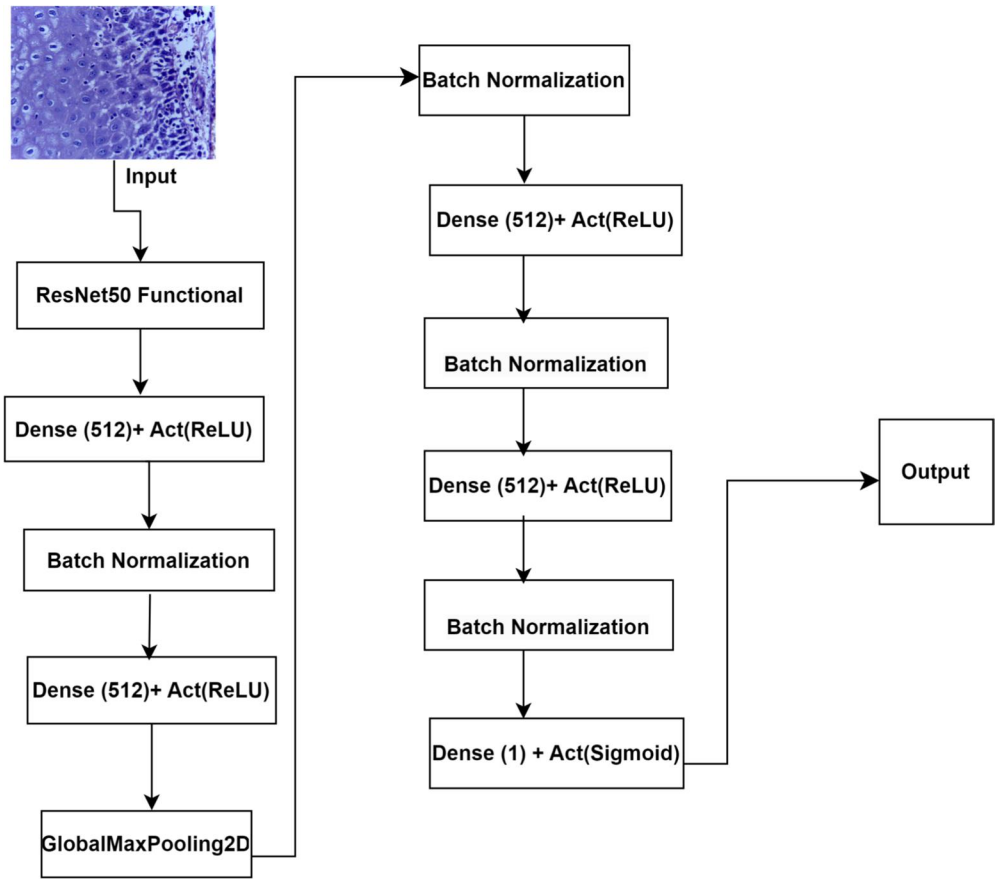

Fig. 2 Architecture of fine-tuned ResNet50

The model development begins with the ResNet50 model, pre-trained on the ImageNet dataset. Initially, the layers of this model are frozen to retain the critical features already learned. This pre-trained model serves as a solid foundation for additional layers for more specific task customization. The additional layers include a dense layer with 512 units and ReLU activation to improve feature extraction, followed by BatchNormalization to stabilize training and prevent overfitting. Another dense layer with 512 units and ReLU activation is added for further feature refinement. This is succeeded by a GlobalMaxPooling2D layer, allowing the model to concentrate on the most significant features, with BatchNormalization again employed for training stability. The architecture is further enriched with two additional dense layers, each having 512 units with ReLU activation, interspersed with BatchNormalization, to optimize the model's learning and interpretive capabilities. The model's final architectural component is a dense output layer with a sigmoid activation function catering to binary classification tasks. This layer is crucial for the model's ability to make precise predictions, as it outputs probabilities that can be interpreted for two distinct classes. Figure 2 represents the fine-tuned ResNet50 architecture.

2.4 Model evaluation and experimentation results

This study assesses the effectiveness of various oversampling and undersampling techniques in addressing class imbalance within histopathology datasets, utilizing a Fine-Tuned ResNet50 model. The training dataset is organized into four subsets designed to reflect varying levels of class imbalance. Each subset contains 171 normal whole slide images. The number of OCSCC whole slide images differs in each subset to mirror the imbalance ratio: subset A includes 17 OCSCC images with an imbalance ratio of 0.1, subset B has 26 OCSCC images with a ratio of 0.15, subset C comprises 35 OCSCC images with a ratio of 0.2, and subset D contains 52 OCSCC images with an imbalance ratio of 0.3. This structure mirrors the class imbalance commonly encountered in medical image datasets. The subsets are processed using oversampling techniques (SMOTE, DeepSMOTE, and ADASYN) and undersampling methods (NearMiss and Edited Nearest Neighbour) to assess their impact on the model's performance. Oversampling increases the number of minority class samples by replicating them or generating synthetic examples, aiming to balance the class distribution. Conversely, undersampling reduces the number

Table 1 Imbalanced ratio after applying oversampling and undersampling

Subsets	Imbalance ratio	Images before sampling			Images after oversampling			Images after undersampling		
		Normal	OCSCC	Total	Normal	OCSCC	Total	Normal	OCSCC	Total
Subset A	0.1	171	17	188	171	171	342	17	17	34
Subset B	0.15	171	26	197	171	171	342	26	26	52
Subset C	0.2	171	35	206	171	171	342	35	35	70
Subset D	0.3	171	52	223	171	171	342	52	52	104

Table 2 Accuracy of fine-tuned ResNet50 model

Subsets	No. of classes	Imbalance ratio	Accuracy				
			SMOTE	Deep SMOTE	ADASYN	NEAR MISS	ENN
Subset A	2	0.1	**0.656**	0.527	0.621	0.542	0.51
Subset B	2	0.15	**0.762**	0.540	0.70	0.5762	0.56
Subset C	2	0.20	**0.830**	0.60	0.762	0.5962	0.64
Subset D	2	0.30	**0.864**	0.661	0.84	0.609	0.661

The bold numbers in the tables demonstrate SMOTE's superior capacity to create synthetic instances compared to other oversampling and undersampling techniques

Table 3 AUC of fine-tuned ResNet50 model

Subsets	No. of classes	Imbalance ratio	AUC				
			SMOTE	Deep SMOTE	ADASYN	Near Miss	ENN
Subset A	2	0.1	**0.662**	0.59	0.61	0.59	0.58
Subset B	2	0.15	**0.874**	0.610	0.762	0.632	0.60
Subset C	2	0.20	**0.890**	0.63	0.782	0.672	0.64
Subset D	2	0.30	**0.918**	0.719	0.84	0.7436	0.711

The bold numbers in the tables demonstrate SMOTE's superior capacity to create synthetic instances compared to other oversampling and undersampling techniques

of majority class samples to achieve a more balanced dataset by removing instances from the majority class. The resulting number of samples after applying oversampling and undersampling is displayed in Table 1.

A fine-tuned ResNet50 architecture is the foundation for evaluating the impact of these techniques. The fine-tuning process, loss function, and hyperparameters, including the training duration (50 epochs), optimizer (Adam), batch size (8), and learning rate (0.001), were meticulously selected to enhance model training efficacy. The model's performance is measured against a testing dataset with standard classification metrics such as accuracy and the Area Under the Curve (AUC). Results, detailed in Tables 2 and 3, indicate that SMOTE significantly outperforms other methods, especially in subset D, where the minority class constitutes 30% of the dataset. Figures 3 and 4 graphically display the results, demonstrating the fluctuations in the performance of each sampling technique as the proportion of OCSCC images in the dataset varies. The Fine-Tuned ResNet50 model exhibits consistent accuracy across the different sampling strategies, demonstrating its robustness. The pronounced success of SMOTE in subset D corroborates its known effectiveness in contexts where the minority class is a substantial portion of the data, a finding supported by existing literature. Such consistency underscores the adaptability of the Fine-Tuned ResNet50 model to a range of sampling strategies, reinforcing its potential application in clinical diagnostic settings.

In addition to pre-trained ResNet50, we evaluated the performance of the pre-trained VGG16 architecture (Simonyan and Zisserman 2014) across the same oversampling and undersampling techniques using different imbalance ratios. The VGG16 model is initialized with weights pre-trained on the ImageNet dataset. Initially, these pre-trained layers are kept frozen to preserve the learned features. To adapt the model for our specific task, we add a dense layer with 512 units and ReLU activation, followed by BatchNormalization to stabilize training. Subsequently, another dense layer with 512 units and ReLU activation is incorporated for further feature refinement. The final layers consist of

Table 4 Accuracy of fine-tuned VGG16 model

Subsets	No. of classes	Imbalance ratio	Accuracy				
			SMOTE	Deep SMOTE	ADASYN	NEAR MISS	ENN
Subset A	2	0.1	**0.625**	0.518	0.592	0.512	0.47
Subset B	2	0.15	**0.683**	0.520	0.672	0.557	0.50
Subset C	2	0.20	**0.719**	0.57	0.70	0.546	0.60
Subset D	2	0.30	**0.745**	0.59	0.735	0.598	0.621

The bold numbers in the tables demonstrate SMOTE's superior capacity to create synthetic instances compared to other oversampling and undersampling techniques

Table 5 AUC of fine-tuned VGG16 model

Subsets	No. of classes	Imbalance ratio	AUC				
			SMOTE	Deep SMOTE	ADASYN	Near Miss	ENN
Subset A	2	0.1	**0.592**	0.55	0.578	0.54	0.50
Subset B	2	0.15	**0.70**	0.568	0.692	0.587	0.52
Subset C	2	0.20	**0.74**	0.60	0.723	0.62	0.63
Subset D	2	0.30	**0.77**	0.64	0.752	0.65	0.66

The bold numbers in the tables demonstrate SMOTE's superior capacity to create synthetic instances compared to other oversampling and undersampling techniques

Table 6 Accuracy of models on 40 percent imbalanced dataset

Models	Imbalance ratio	Accuracy				
		SMOTE	Deep SMOTE	ADASYN	Near Miss	ENN
ResNet	0.40	**0.89**	0.68	0.86	0.62	0.68
VGG16	0.40	**0.77**	0.62	0.77	0.60	0.64

The bold numbers in the tables demonstrate SMOTE's superior capacity to create synthetic instances compared to other oversampling and undersampling techniques

a GlobalMaxPooling2D layer for feature concentration and four additional dense layers with 512 units each, interspersed with BatchNormalization, to enhance learning and interpretability. The model concludes with a dense output layer using a sigmoid activation function for binary

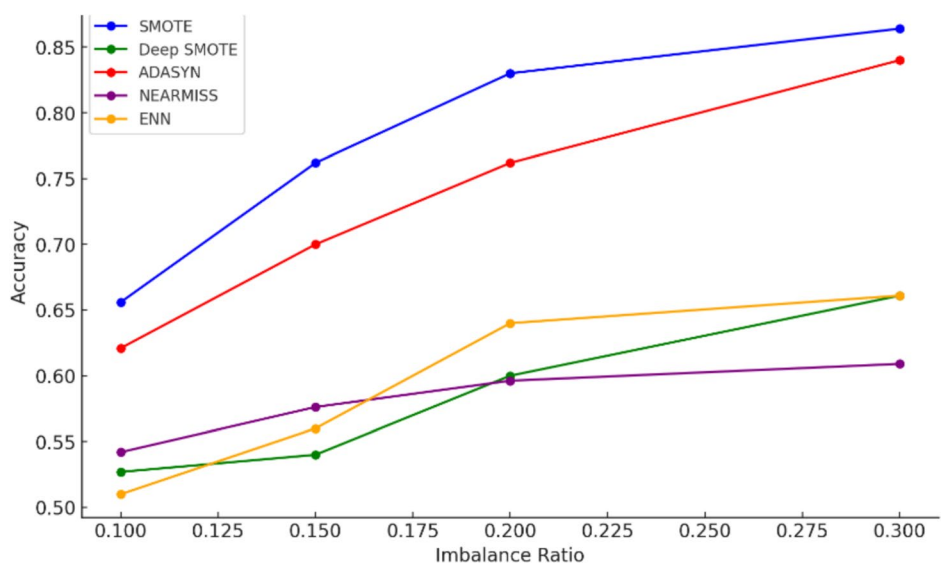

Fig. 3 Accuracy comparison of sampling techniques on histopathological OCSCC dataset by varying the imbalance ratio

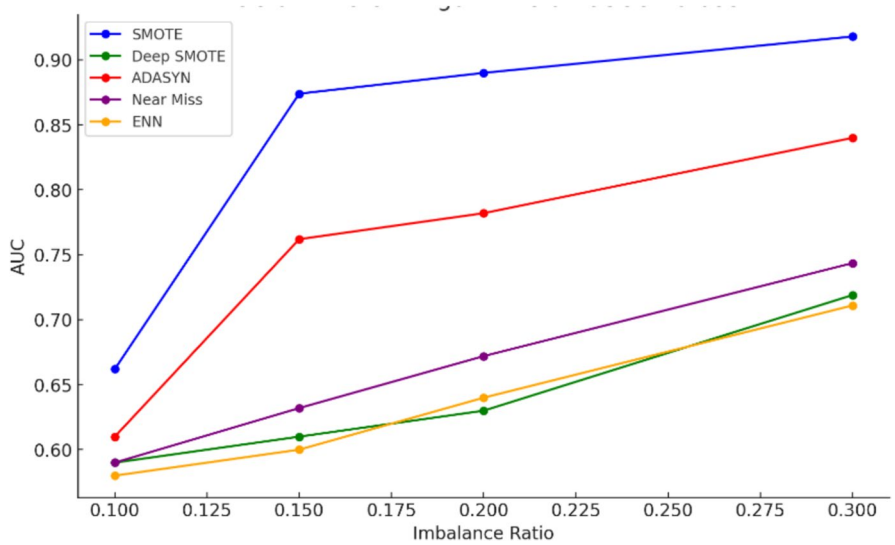

Fig. 4 AUC Comparison of sampling techniques on histopathological OCSCC dataset by varying the imbalance ratio

classification tasks. The accuracy and Auc result of Vgg16 are displaced in Tables 4 and 5.

3 Discussion

Accurately classifying histopathological images such as Oral Squamous Cell Carcinoma (OSCC) and normal tissues is crucial for effective patient care. However, imbalances within the datasets used for training deep learning models significantly complicate this task, degrading the accuracy of models, especially in detecting OSCC from histopathology images. Our study addresses this critical issue of class imbalance, aiming to enhance the precision and reliability of diagnostic models by tackling the disparities in data distribution. Our research confirms that the strategic implementation of oversampling (SMOTE, DeepSMOTE, and ADASYN) and undersampling techniques (NearMiss and Edited Nearest Neighbor), particularly SMOTE, considerably enhances the performance of these models. This is especially true in scenarios of class imbalance, as demonstrated by the superior performance of the fine-tuned ResNet50 model, which achieves an accuracy of 0.864 and an AUC of 0.918 in the imbalanced subset D, and by the VGG16 model, which also sees performance gains with accuracy and AUC scores of 0.745 and 0.77, respectively. Our study reveals SMOTE as the standout technique for addressing class imbalance, with its effectiveness reflected across various subsets, as seen in Tables 2, 3, 4, and 5. These results showcase SMOTE's capability to create synthetic instances that strengthen the model's generalization, thereby improving OSCC classification performance, as indicated by the bold numbers in the tables.

Further integrating transfer learning with the SMOTE algorithm enhances these results, utilizing the advanced capabilities of pre-trained models like ResNet50. ResNet50's architectural intricacies, such as residual connections, help overcome the vanishing gradient problem, facilitating more practical training on deep architectures. This contrasts with the VGG16 model, which, lacking these architectural features, tends to underperform in deeper networks. Such insights are evidenced by ResNet50's consistent outperformance across all imbalance ratios examined. The robustness of the models is further evaluated by reducing the class imbalance and increasing the proportion of OSCC cases to a 40% ratio. Interestingly, both models demonstrate improved performance at this less imbalanced ratio compared to all previously tested ratios, as evidenced by the results in the Table 6. However, the results show only minor differences when comparing performance with other ratios at 40% data. This indicates that oversampling techniques effectively balance the data, allowing both models to achieve improved performance.

4 Conclusion

This study significantly advances the automated diagnosis of Oral Cavity Squamous Cell Carcinoma (OCSCC) using histopathological images. Our research study demonstrates the substantial impact of sophisticated oversampling and undersampling techniques by addressing class imbalance in medical datasets, particularly in OCSCC. Employing various imbalance ratios within the publicly available OCSCC dataset, we have explored the efficacy of SMOTE, Deep SMOTE, ADASYN, Near Miss, and Edited Nearest Neighbors in conjunction with transfer learning. The results reveal that the strategic application of SMOTE with transfer learning significantly improves the classification accuracy of 86% and AUC of 0.91. By automating and optimizing the diagnostic process, this research substantially contributes to efficient patient treatment, enhanced outcomes, and reduced diagnostic expenses in oral cancer

diagnosis, marking a remarkable advancement in medical technology and patient care. Apart from the impressive performance demonstrated by the models in this study, there is significant potential for further enhancements and refinements in their capabilities for automated detection of oral cavity squamous cell carcinoma. A promising direction for future exploration involves advanced data augmentation techniques and the integration of Transformer-based mechanisms.

Acknowledgements We thank our host institution, the Islamic University of Science and Technology (IUST), for providing us with High-Performance Computing (HPC) resources and the technical team of IUST Cloud for their assistance.

Funding The authors declare that this research was supported by the project "Artificial Intelligence-based Deep Learning Model to Predict Human Papilloma Virus Infection in Head and Neck Squamous Cell Carcinoma Patients Using Histopathology Image", funded by the Jammu and Kashmir Science and Technology Department (Grant No. JKST&IC/SRE/926-30, dated 29/01/2024), and the Promotion of University Research and Scientific Excellence (PURSE) project (Grant No. SR/PURSE/2022/121), funded by the Department of Science and Technology, Government of India, New Delhi to the Islamic University of Science and Technolgy (IUST), Awantipora.

Declarations

Conflict of interest The authors affirm that there are no Conflict of interest related to the publication of this paper.

Ethics approval This paper does not include any research involving human participants or animals conducted by the authors.

References

Altaf F, Islam SM, Akhtar N, Janjua NK (2019) Going deep in medical image analysis: concepts, methods, challenges, and future directions. IEEE Access 7:99540–99572

Anwar N, Pervez S, Chundriger Q, Awan S, Moatter T, Ali TS (2020) Oral cancer: clinicopathological features and associated risk factors in a high risk population presenting to a major tertiary care center in Pakistan. PLoS ONE 15(8):0236359

Chakraborty D, Natarajan C, Mukherjee A (2019) Advances in oral cancer detection. Adv Clin Chem 91:181–200

Chawla NV, Bowyer KW, Hall LO, Kegelmeyer WP (2002) Smote: synthetic minority over-sampling technique. J Artif Intell Res 16:321–357

Choi S, Kim YJ, Briceno S, Mavris D (2016) Prediction of weather-induced airline delays based on machine learning algorithms. In: 2016 IEEE/AIAA 35th Digital Avionics Systems Conference (DASC). IEEE, pp 1–6

Dablain D, Krawczyk B, Chawla NV (2022) Deepsmote: Fusing deep learning and smote for imbalanced data. IEEE Transactions on Neural Networks and Learning Systems

Das DK, Bose S, Maiti AK, Mitra B, Mukherjee G, Dutta PK (2018) Automatic identification of clinically relevant regions from oral tissue histological images for oral squamous cell carcinoma diagnosis. Tissue Cell 53:111–119

Deif MA, Solyman AA, Alsharif MH, Uthansakul P (2021) Automated triage system for intensive care admissions during the covid-19 pandemic using hybrid xgboost-ahp approach. Sensors 21(19):6379

Du M, Nair R, Jamieson L, Liu Z, Bi P (2020) Incidence trends of lip, oral cavity, and pharyngeal cancers: global burden of disease 1990–2017. J Dent Res 99(2):143–151

Duggento A, Conti A, Mauriello A, Guerrisi M, Toschi N (2021) Deep computational pathology in breast cancer. In: Seminars in Cancer Biology, 72. Elsevier, pp 226–237

Echle A, Rindtorff NT, Brinker TJ, Luedde T, Pearson AT, Kather JN (2021) Deep learning in cancer pathology: a new generation of clinical biomarkers. Br J Cancer 124(4):686–696

Folmsbee J, Liu X, Brandwein-Weber M, Doyle S (2018) Active deep learning: Improved training efficiency of convolutional neural networks for tissue classification in oral cavity cancer. In: 2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI 2018). IEEE, pp 770–773

Goldenberg SL, Nir G, Salcudean SE (2019) A new era: artificial intelligence and machine learning in prostate cancer. Nat Rev Urol 16(7):391–403

He H, Bai Y, Garcia EA, Li S (2008) Adasyn: Adaptive synthetic sampling approach for imbalanced learning. In: 2008 IEEE International Joint Conference on Neural Networks (IEEE World Congress on Computational Intelligence). Ieee, pp 1322–1328

He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp 770–778

Li L, Yin Y, Nan F, Ma Z (2022) Circ_lpar3 promotes the progression of oral squamous cell carcinoma (oscc). Biochem Biophys Res Commun 589:215–222

Liu J-E, An F-P (2020) Image classification algorithm based on deep learning-kernel function. Sci Program 2020:1–14

McCann MT, Ozolek JA, Castro CA, Parvin B, Kovacevic J (2014) Automated histology analysis: opportunities for signal processing. IEEE Signal Process Mag 32(1):78–87

Mittal U, Srivastava S, Chawla P (2019) Review of different techniques for object detection using deep learning. In: Proceedings of the Third International Conference on Advanced Informatics for Computing Research, pp 1–8

Santana MF, Ferreira LCL (2017) Diagnostic errors in surgical pathology. Jornal Brasileiro de Patologia e Medicina Laboratorial 53:124–129

Simonyan K, Zisserman A (2014) Very deep convolutional networks for large-scale image recognition. arXiv preprint arXiv:1409.1556

Wang S, Yang DM, Rong R, Zhan X, Fujimoto J, Liu H, Minna J, Wistuba II, Xie Y, Xiao G (2019) Artificial intelligence in lung cancer pathology image analysis. Cancers 11(11):1673

Warnakulasuriya S, Greenspan JS (2020) Epidemiology of oral and oropharyngeal cancers. In: Textbook of Oral Cancer. Springer, pp 5–21

Wei W, Li J, Cao L, Ou Y, Chen J (2013) Effective detection of sophisticated online banking fraud on extremely imbalanced data. World Wide Web 16:449–475

Wilson DL (1972) Asymptotic properties of nearest neighbor rules using edited data. IEEE Trans Syst, Man, Cybern 3:408–421

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.